

4 Weierstrass infinite products

We now turn to Weierstrass's construction of an entire function with prescribed zeros.

Theorem 4.1 *Given any sequence $\{a_n\}$ of complex numbers with $|a_n| \rightarrow \infty$ as $n \rightarrow \infty$, there exists an entire function f that vanishes at all $z = a_n$ and nowhere else. Any other such entire function is of the form $f(z)e^{g(z)}$, where g is entire.*

Recall that if a holomorphic function f vanishes at $z = a$, then the multiplicity of the zero a is the integer m so that

$$f(z) = (z - a)^m g(z),$$

where g is holomorphic and nowhere vanishing in a neighborhood of a . Alternatively, m is the first non-zero power of $z - a$ in the power series expansion of f at a . Since, as before, we allow for repetitions in the sequence $\{a_n\}$, the theorem actually guarantees the existence of entire functions with prescribed zeros and with desired multiplicities.

To begin the proof, note first that if f_1 and f_2 are two entire functions that vanish at all $z = a_n$ and nowhere else, then f_1/f_2 has removable singularities at all the points a_n . Hence f_1/f_2 is entire and vanishes nowhere, so that there exists an entire function g with $f_1(z)/f_2(z) = e^{g(z)}$, as we showed in Section 6 of Chapter 3. Therefore $f_1(z) = f_2(z)e^{g(z)}$ and the last statement of the theorem is verified.

Hence we are left with the task of constructing a function that vanishes at all the points of the sequence $\{a_n\}$ and nowhere else. A naive guess, suggested by the product formula for $\sin \pi z$, is the product $\prod_n (1 - z/a_n)$. The problem is that this product converges only for suitable sequences $\{a_n\}$, so we correct this by inserting exponential factors. These factors will make the product converge without adding new zeros.

For each integer $k \geq 0$ we define **canonical factors** by

$$E_0(z) = 1 - z \quad \text{and} \quad E_k(z) = (1 - z)e^{z + z^2/2 + \cdots + z^k/k}, \quad \text{for } k \geq 1.$$

The integer k is called the **degree** of the canonical factor.

Lemma 4.2 *If $|z| \leq 1/2$, then $|1 - E_k(z)| \leq c|z|^{k+1}$ for some $c > 0$.*

Proof. If $|z| \leq 1/2$, then with the logarithm defined in terms of the power series, we have $1 - z = e^{\log(1-z)}$, and therefore

$$E_k(z) = e^{\log(1-z) + z + z^2/2 + \cdots + z^k/k} = e^w,$$

where $w = -\sum_{n=k+1}^{\infty} z^n/n$. Observe that since $|z| \leq 1/2$ we have

$$|w| \leq |z|^{k+1} \sum_{n=k+1}^{\infty} |z|^{n-k-1}/n \leq |z|^{k+1} \sum_{j=0}^{\infty} 2^{-j} \leq 2|z|^{k+1}.$$

In particular, we have $|w| \leq 1$ and this implies that

$$|1 - E_k(z)| = |1 - e^w| \leq c'|w| \leq c|z|^{k+1}.$$

Remark. An important technical point is that the constant c in the statement of the lemma can be chosen to be independent of k . In fact, an examination of the proof shows that we may take $c' = e$ and then $c = 2e$.

Suppose that we are given a zero of order m at the origin, and that $a_1, a_2, \dots$ are all non-zero. Then we define the Weierstrass product by

$$f(z) = z^m \prod_{n=1}^{\infty} E_n(z/a_n).$$

We claim that this function has the required properties; that is, f is entire with a zero of order m at the origin, zeros at each point of the sequence $\{a_n\}$, and f vanishes nowhere else.

Fix $R > 0$, and suppose that z belongs to the disc $|z| < R$. We shall prove that f has all the desired properties in this disc, and since R is arbitrary, this will prove the theorem.

We can consider two types of factors in the formula defining f , with the choice depending on whether $|a_n| \leq 2R$ or $|a_n| > 2R$. There are only finitely many terms of the first kind (since $|a_n| \rightarrow \infty$), and we see that the finite product vanishes at all $z = a_n$ with $|a_n| < R$. If $|a_n| \geq 2R$, we have $|z/a_n| \leq 1/2$, hence the previous lemma implies

$$|1 - E_n(z/a_n)| \leq c \left| \frac{z}{a_n} \right|^{n+1} \leq \frac{c}{2^{n+1}}.$$

Note that by the above remark, c does not depend on n . Therefore, the product

$$\prod_{|a_n| \geq 2R} E_n(z/a_n)$$

defines a holomorphic function when $|z| < R$, and does not vanish in that disc by the propositions in Section 3. This shows that the function

f has the desired properties, and the proof of Weierstrass's theorem is complete.

5 Hadamard's factorization theorem

The theorem of this section combines the results relating the growth of a function to the number of zeros it possesses, and the above product theorem. Weierstrass's theorem states that a function that vanishes at the points $a_1, a_2, \dots$ takes the form

$$e^{g(z)} z^m \prod_{n=1}^{\infty} E_n(z/a_n).$$

Hadamard refined this result by showing that in the case of functions of finite order, the degree of the canonical factors can be taken to be constant, and g is then a polynomial.

Recall that an entire function has an order of growth $\leq \rho$ if

$$|f(z)| \leq A e^{B|z|^\rho},$$

and that *the* order of growth ρ_0 of f is the infimum of all such ρ 's.

A basic result we proved earlier was that if f has order of growth $\leq \rho$, then

$$n(r) \leq C r^\rho, \quad \text{for all large } r,$$

and if $a_1, a_2, \dots$ are the non-zero zeros of f , and $s > \rho$, then

$$\sum |a_n|^{-s} < \infty.$$

Theorem 5.1 *Suppose f is entire and has growth order ρ_0 . Let k be the integer so that $k \leq \rho_0 < k + 1$. If $a_1, a_2, \dots$ denote the (non-zero) zeros of f , then*

$$f(z) = e^{P(z)} z^m \prod_{n=1}^{\infty} E_k(z/a_n),$$

where P is a polynomial of degree $\leq k$, and m is the order of the zero of f at $z = 0$.

Main lemmas

Here we gather a few lemmas needed in the proof of Hadamard's theorem.